(1) $\lim\limits_{n\to\infty}(1+x_n)^{y_n}$ Trudniejsze

(2) lim z R.H ← de Hospitala *

(3) znaleźć obszar zb. szeregu „z x"

(4) pochodna skomplikowanej fali złożonej

(5) Badanie przebiegu zmienności funkcji ln *

W. Stankiewicz – zbiór zadań

Stosując R.H. oblicz

a)

$$\lim_{x\to 0}\frac{x-\sin x}{x-\operatorname{tg} x}=\left[\frac{0-0}{0-0}=\frac{0}{0}\right]\to\frac{1-\cos x}{1-\frac{1}{\cos^2 x}}\to\frac{(1-\cos x)\cos^2 x}{\cos^2 x-1}\to$$

$$\to\frac{-(\cancel{\cos x-1})\cos^2 x}{(\cancel{\cos x-1})(\cos x+1)}\to\frac{-\cos^2 x}{\cos x+1}\to-\frac{1}{2}$$

b)

$$\lim_{x\to\infty}\frac{\arcsin\frac{1}{x}}{\operatorname{arcctg} x-\frac{\pi}{2}}=\left[\frac{0}{\frac{\pi}{2}-\frac{\pi}{2}}=\frac{0}{0}\right]\xrightarrow{RH}\frac{\frac{1}{\sqrt{1-\frac{1}{x^2}}}\cdot\left(-\frac{1}{x^2}\right)}{\frac{1}{1+x^2}}\to$$

$$\to\frac{\frac{-1}{\sqrt{x^4-x^2}}}{\frac{1}{1+x^2}}\to\frac{-(x^2+1)}{x\sqrt{x^2-1}}\to\frac{-x^2-1}{x^2\sqrt{1-\frac{1}{x^2}}}$$

$$\frac{\overset{1}{\cancel{x^2}}\left(-1-\overset{0}{\frac{1}{x^2}}\right)}{\underset{1}{\cancel{x^2}}\sqrt{1-\frac{1}{x^2}}}\to-1$$

c)

$$\lim_{x\to 1^-}\frac{\ln(1-x)}{\ln(\sin(1-x))}=\left[\frac{-\infty}{-\infty}\right]\xrightarrow{RH}\frac{\frac{-\cancel{1}\,1}{1-x}}{\frac{1}{\sin(1-x)}\cdot\cos(1-x)\cdot\cancel{(-1)}\,1}\to$$

$$\to\frac{\frac{1}{1-x}}{\operatorname{ctg}(1-x)}\to\frac{\operatorname{tg}(1-x)}{1-x}\to\left[\frac{0^-}{0^-}\right]\xrightarrow{RH}\frac{\frac{-1}{\cos^2(1-x)}}{-1}\to$$

$$\to\frac{1}{\cos^2(1-x)}\to 1$$

d)

$$\lim_{x\to 0^-}(x\ln x)=[0\cdot(-\infty)]\to\frac{\ln x}{\frac{1}{x}}\to\left[\frac{-\infty}{\infty}\right]\xrightarrow{RH}\frac{\frac{1}{x}}{-\frac{1}{x^2}}\to -x\to 0$$

e)

$$\lim_{x\to 1^-}\ln x\cdot\ln(1-x)=[0\cdot(-\infty)]\to\frac{\ln(1-x)}{\frac{1}{\ln x}}=\left[\frac{-\infty}{\infty}\right]\xrightarrow{RH}$$

$$\to\frac{\frac{\cancel{-}1}{1-x}}{\frac{\cancel{-}1}{\ln^2 x}\cdot\frac{1}{x}}\to\frac{\ln^2 x\cdot x}{1-x}\to\left[\frac{0}{0}\right]\to$$

$$\to\frac{1\cdot\overset{0}{\ln^2 x}+2\overset{0}{\ln x}\cdot\overset{1}{\frac{1}{x}}}{-1}\to\left[\frac{0}{-1}\right]$$

$$\to 0$$

f)

$$\lim_{x\to\infty}xe^{-5x}=[\infty\cdot 0]\to\frac{x}{e^{5x}}\to\left[\frac{\infty}{\infty}\right]\xrightarrow{RH}\frac{1}{5e^{5x}}\to 0$$

g)

$$\lim_{x\to\infty}\left((\pi-2\operatorname{arctg}x)\ln x\right)=\left[\left(\pi-2\cdot\frac{\pi}{2}\right)\cdot\infty=0\cdot\infty\right]\to$$

$$\to\frac{\pi-2\operatorname{arctg}x}{\frac{1}{\ln x}}\to\left[\frac{0}{0}\right]\xrightarrow{RH}\frac{\frac{\cancel{-}2}{1+x^2}}{\frac{\cancel{-}1}{\ln^2 x}\cdot\frac{1}{x}}\to\frac{2\ln^2 x\cdot x}{1+x^2}\to\left[\frac{\infty}{\infty}\right]\to$$

$$\xrightarrow{RH}\frac{\overset{1}{\cancel{2}}\,\overset{1}{\cancel{x}}\cdot 2\ln x\cdot\frac{1}{\overset{1}{\cancel{x}}}+\overset{1}{\cancel{2}}\ln^2 x}{\underset{1}{\cancel{2}}x}\to\frac{2\ln x+\ln^2 x}{x}\xrightarrow{RH}\frac{2\cdot\ln x\cdot\frac{1}{x}+2\cdot\frac{1}{x}}{1}\to$$

$$\to\frac{2(\ln x+1)}{x}\xrightarrow{H}\frac{2\cdot\overset{\nearrow 0}{\frac{1}{x}}}{1}\to 0$$

h)

$$\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)=\left[\frac{1}{0}-\frac{1}{0}=\infty-\infty\right]\to\frac{x\cdot\ln x-x+1}{\ln x\,(x-1)}\to\left[\frac{1\cdot 0-1+1}{0\cdot(1-1)}=\frac{0}{0}\right]=$$

$$\xrightarrow{RH}\frac{\ln x+\frac{1}{x}\cdot x-1}{\frac{1}{x}(x-1)+\ln x}\to\frac{\ln x}{\ln x+\frac{x-1}{x}}\to\left[\frac{0}{0+0}\right]\xrightarrow{RH}\frac{\frac{1}{x}}{\frac{1}{x}+\frac{x-(x-1)}{x^2}}\to$$

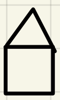
$$\to\frac{\frac{1}{x}}{\frac{1}{x}+\frac{1}{x^2}}\to\frac{1}{2}$$

i)

$$\lim_{x\to\infty}\left((x+2)\overset{\nearrow 1}{e^{\overset{\nearrow 0}{\frac{1}{x}}}}-x\right)=[\infty-\infty]\to x\left(\frac{x+2}{x}e^{\frac{1}{x}}-1\right)\to[\infty\cdot 0]\to$$

$$\to\frac{\frac{x+2}{x}\cdot e^{\frac{1}{x}}-1}{\frac{1}{x}}\to\left[\frac{0}{0}\right]\to$$

$$\to\frac{\left(1+\frac{2}{x}\right)\cdot e^{\frac{1}{x}}-1}{\frac{1}{x}}\to\left[\frac{0}{0}\right]\xrightarrow{H}\frac{\cancel{-}\frac{2}{x^2}e^{\frac{1}{x}}+\left(1+\frac{2}{x}\right)\cdot\cancel{-}\frac{1}{x^2}\cdot e^{\frac{1}{x}}}{\cancel{-}\frac{1}{x^2}}\to$$



$$\to\frac{e^{\frac{1}{x}}\left(\frac{2}{x^2}+\frac{1}{x^2}+\frac{2}{x^3}\right)}{\frac{1}{x^2}}\to$$

25.23 d f

25.24 e f

25.25 d h k

$$\to\underset{\uparrow 1}{e^{\frac{1}{x}}}\left(3+\overset{\nearrow 0}{\frac{2}{x}}\right)\to 3$$

$f^g=e^{g\ln f}$

j)

$$\lim_{x\to 0^+}x^{x^2}=[0^0]=\lim_{x\to 0^+}e^{x^2\ln x}\to e^0\to 1$$

$$\lim_{x\to 0^+}x^2\cdot\ln x=[0\cdot-\infty]\to\frac{\ln x}{\frac{1}{x^2}}\to\left[\frac{0}{0}\right]\xrightarrow{H}\frac{\frac{1}{x}}{\frac{-2}{x^3}}\to\frac{1}{x}\cdot\frac{x^3}{-2}\to\frac{x^2}{-2}\to 0$$

$$\lim_{x\to 0^+} (\arcsin x)^{\sin x} = [0^0] = \lim_{x\to 0^+} e^{\sin x \cdot \ln(\arcsin x)}$$

$$\lim_{x\to 0^+} \sin x \cdot \ln(\arcsin x) \to [0 \cdot (-\infty)] \to \frac{\ln(\arcsin x)}{\frac{1}{\sin x}} \to \left[\frac{-\infty}{\infty}\right] \to$$

$$\to \frac{\frac{1}{\arcsin x} \cdot \frac{1}{\sqrt{1-x^2}}}{\frac{-1}{\sin^2 x} \cdot \cos x} \to \frac{-\sin x \cdot tg\, x}{\arcsin x \cdot \sqrt{1-x^2}} \to$$

$$\to \left[\frac{0}{0}\right] \xrightarrow{RH} \frac{-\left(\cos x \cdot tg\, x - \sin x \frac{1}{\cos^2 x}\right)}{\left(\frac{1}{\sqrt{\underset{1}{\cancel{1-x^2}}}} \cdot \sqrt{\underset{1}{\cancel{1-x^2}}} - \arcsin x \frac{-2x}{2\sqrt{1-x^2}}\right.}$$

$$\to \frac{-\sin x + \sin x \frac{1}{\cos^2 x}}{(1 -}$$